

Praneel Magapu

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Education

North Carolina State University
B.S. Computer Science, GPA 3.5 / 4.0

- Independent research studies: CSC 498 (retrieval-augmented teaching assistant, Game2Learn Lab) and CSC 490 (crowd-label quality control). Robotics Track certificate coursework in progress.

Aug 2023 – May 2027 (expected)
Raleigh, NC

Experience

Align Technology
Software Engineering Intern, Product R&D

- Redesigned support-structure generation inside a large production C++ pipeline for directly 3D-printed medical devices, improving surface quality while reducing downstream manual finishing and material use.
- Implemented mesh-processing algorithms (Eigen, Open3D) that derive valid placement regions from part geometry so the system generalizes across device shapes; shipped into an established codebase under a regulated medical-device development process.

May – Aug 2026
Raleigh, NC

DevDynamics.ai
Software Engineering Intern

- Built the backend intelligence layer for InboxFlow, a Gmail workflow service that classifies transactional email and tracks order, return, and refund state across a thread.
- Designed a two-stage classifier that runs deterministic regex on the fast path and escalates only low-confidence or unrecognized messages to a Gemini LLM, holding down inference cost and latency while still catching phrasing that pattern matching misses.
- Mapped structured LLM output (event type, vendor, primary value, summary, line items) back into the existing fixed label schema, so every downstream consumer stays agnostic to which classification path produced the result.
- Migrated thread-timeline persistence from an in-memory store to Supabase (PostgreSQL), resolving a primary-key sequence defect by moving the schema to auto-generated BIGSERIAL IDs, and deployed the Flask service to AWS Elastic Beanstalk as a stable public endpoint for a Gmail Add-on.

May – Aug 2025
Remote

Liquid Rocketry Lab, NC State
Data Engineer

- Built the real-time telemetry pipeline for live liquid rocket engine hot-fire tests in Python and MongoDB across containerized microservices, capturing abort and breach events at sub-second latency.
- Designed an observable fault-detection pattern that decouples threshold logic from the ingestion path, so new fault conditions ship by subscribing a detector rather than editing the pipeline.

Sep 2024 – May 2025
Raleigh, NC

Research

NC State Center for Geospatial Analytics
Undergraduate Research Assistant, GAEC Lab

- Extend a harmonized flood-segmentation training pipeline by integrating five published benchmark corpora, reconciling conflicting coordinate reference systems, resolutions, sensor modalities, and label taxonomies.
- Contribute to FloodAtlas, fusing Sentinel-1 SAR with optical imagery to detect ephemeral flooding at global scale; selected for the I-GROUP funded research appointment.

Aug 2026 – Present
Raleigh, NC

Projects

Hiring Agent | Python, FastAPI, React, Docker, GitHub Actions, Pydantic

- Built a resume-to-job-description matching pipeline with hybrid scoring that combines deterministic keyword matching with LLM semantic evaluation, producing a written rationale alongside every score.
- Enforced typed, validated contracts on LLM output with Pydantic models, rejecting malformed responses at the API boundary instead of letting them propagate into application logic; implemented ontology-based skill normalization.
- Containerized with Docker and wired a GitHub Actions CI pipeline with pytest coverage.

2026

Mendacia | Python, Flask, React, TypeScript, TwelveLabs, LLM APIs

- Built a full-stack multimodal media-forensics platform that turns raw video into a structured, explainable report flagging propaganda techniques and cross-modal inconsistencies between LLM-extracted claims and detected scene content.
- Designed a defensive adapter layer that normalizes backend responses into a strict UI schema before they reach application state.

Hack_NCState 2026

Technical Skills

Languages: Python, C++ (C++17), Java, SQL, TypeScript, JavaScript, Bash
Backend & Web: Flask, FastAPI, Pydantic, REST API design, React
Data & Cloud: PostgreSQL, Supabase, MongoDB, AWS (Elastic Beanstalk, CLI), Docker, GitHub Actions, Jenkins, CI/CD
AI: LLM integration and structured output, RAG systems, Gemini API